Causal Analysis Report

bpic12-a.xes.gz-dfg

Decision points:	30582
Choice sets:	4
Transitions fitted:	12
Significant (p<0.05):	0
Generated:	2026-06-10 08:47

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

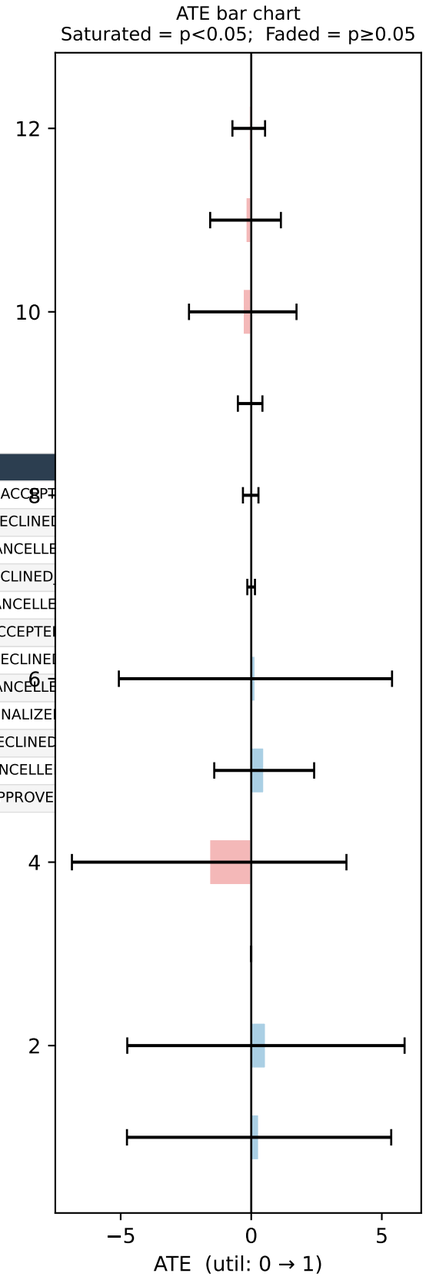
choice_set	choice_set_labels	n	choices_seen	coef	ci_95	perm_p_value	reject_H0	note
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	1.7696	[-2.629, 2.016]	0.0	True	higher util → t3 (A_CANCELLED_t3) over t1 (A_PREACCEPTED_t1)
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	-3.4801	[-4.049, -2.922]	0.0	True	higher util → t4 (A_DECLINED_t4) over t6 (A_ACCEPTED_t6)
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	-0.5457	[-1.558, 0.118]	0.05	False	higher util → t7 (A_DECLINED_t7) over t9 (A_FINALIZED_t9)
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	-5.7161	[-6.138, -5.304]	0.0	True	higher util → t10 (A_DECLINED_t10) over t12 (A_APPROVED_t12)

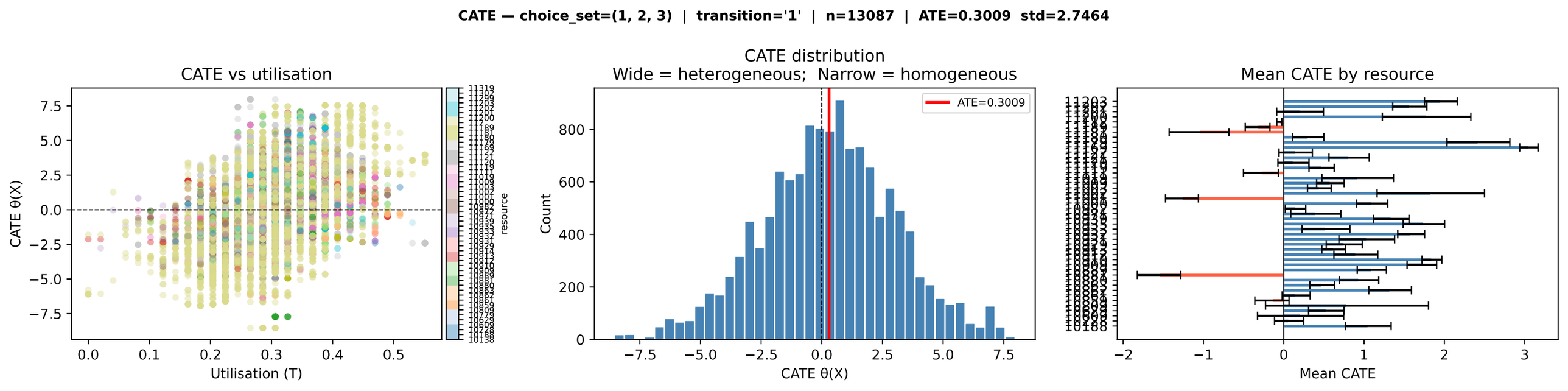
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

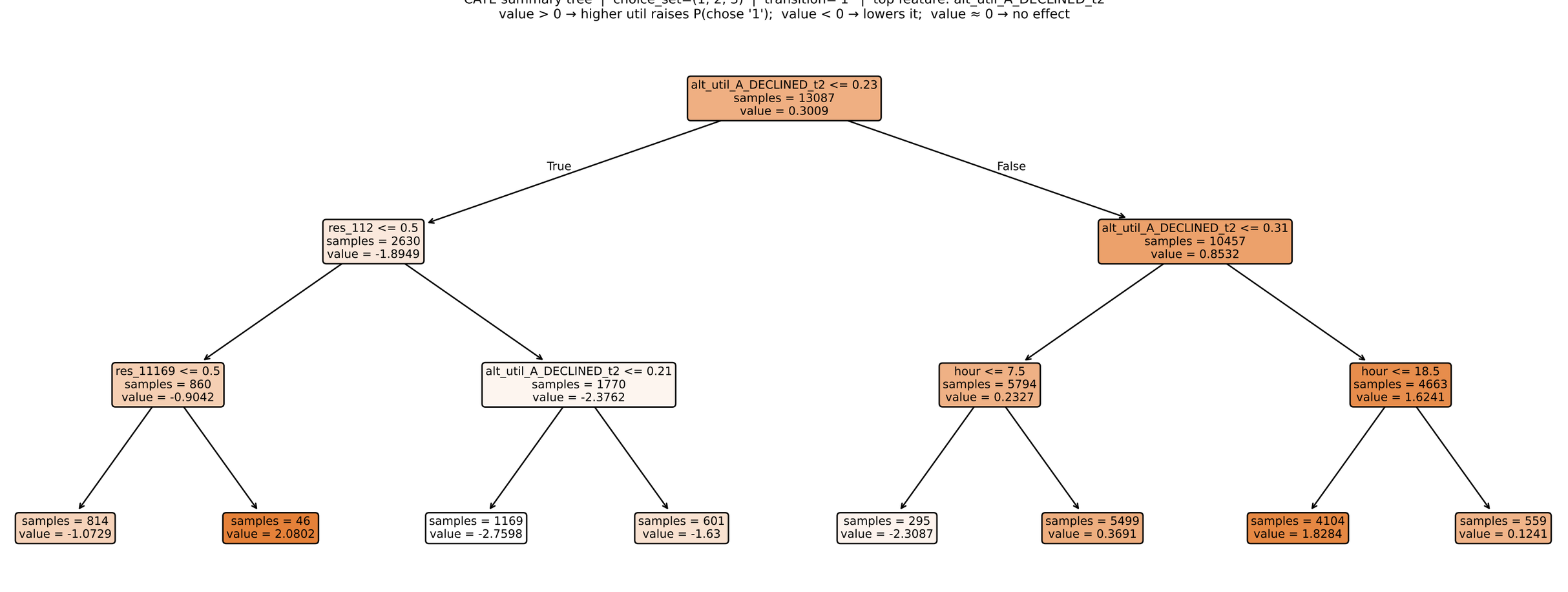
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	1	A_PREACCEPTED_t1	A_CANCELLED_t3, A_DECLINED_t2	0.390798	0.303825	[-0.2047, 0.9863]	0.1984	False	θ=+0.3908: 1-unit ↑ util ↑ P(t1/A_PREACCEPTED_t1) by 39.1%
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	2	A_DECLINED_t2	A_CANCELLED_t3, A_PREACCEPTED_t1	0.75528	0.312298	[0.1432, 1.3674]	0.0156	True	θ=+0.7553: 1-unit ↑ util ↑ P(t2/A_DECLINED_t2) by 75.5%
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	3	A_CANCELLED_t3	A_DECLINED_t2, A_PREACCEPTED_t1	-6.1e-05	8.1e-05	[-0.0002, 0.0001]	0.4535	False	θ=-0.0001: 1-unit ↑ util ↓ P(t3/A_CANCELLED_t3) by 0.0%
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	4	A_DECLINED_t4	A_ACCEPTED_t6, A_CANCELLED_t5	-1.242285	0.405757	[-2.0376, -0.4470]	0.0022	True	θ=-1.2423: 1-unit ↑ util ↓ P(t4/A_DECLINED_t4) by 124.2%
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	5	A_CANCELLED_t5	A_ACCEPTED_t6, A_DECLINED_t4	0.418555	0.158019	[0.1088, 0.7283]	0.0081	True	θ=+0.4186: 1-unit ↑ util ↑ P(t5/A_CANCELLED_t5) by 41.9%
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	6	A_ACCEPTED_t6	A_CANCELLED_t5, A_DECLINED_t4	0.34102	0.392376	[-0.4280, 1.1101]	0.3848	False	θ=+0.3410: 1-unit ↑ util ↑ P(t6/A_ACCEPTED_t6) by 34.1%
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	7	A_DECLINED_t7	A_CANCELLED_t8, A_FINALIZED_t9	0.000451	0.012026	[-0.0231, 0.0240]	0.9701	False	θ=+0.0005: 1-unit ↑ util ↑ P(t7/A_DECLINED_t7) by 0.0%
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	8	A_CANCELLED_t8	A_DECLINED_t7, A_FINALIZED_t9	-0.027322	0.026872	[-0.0800, 0.0253]	0.3093	False	θ=-0.0273: 1-unit ↑ util ↓ P(t8/A_CANCELLED_t8) by 2.7%
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	9	A_FINALIZED_t9	A_CANCELLED_t8, A_DECLINED_t7	-0.044183	0.043515	[-0.1295, 0.0411]	0.3099	False	θ=-0.0442: 1-unit ↑ util ↓ P(t9/A_FINALIZED_t9) by 4.4%
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	10	A_DECLINED_t10	A_APPROVED_t12, A_CANCELLED_t11	-0.25615	0.242839	[-0.7321, 0.2198]	0.2915	False	θ=-0.2561: 1-unit ↑ util ↓ P(t10/A_DECLINED_t10) by 25.6%
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	11	A_CANCELLED_t11	A_APPROVED_t12, A_DECLINED_t10	-0.129325	0.136757	[-0.3974, 0.1387]	0.3443	False	θ=-0.1293: 1-unit ↑ util ↓ P(t11/A_CANCELLED_t11) by 12.9%
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	12	A_APPROVED_t12	A_CANCELLED_t11, A_DECLINED_t10	0.031104	0.073209	[-0.1124, 0.1746]	0.6709	False	θ=+0.0311: 1-unit ↑ util ↑ P(t12/A_APPROVED_t12) by 3.1%

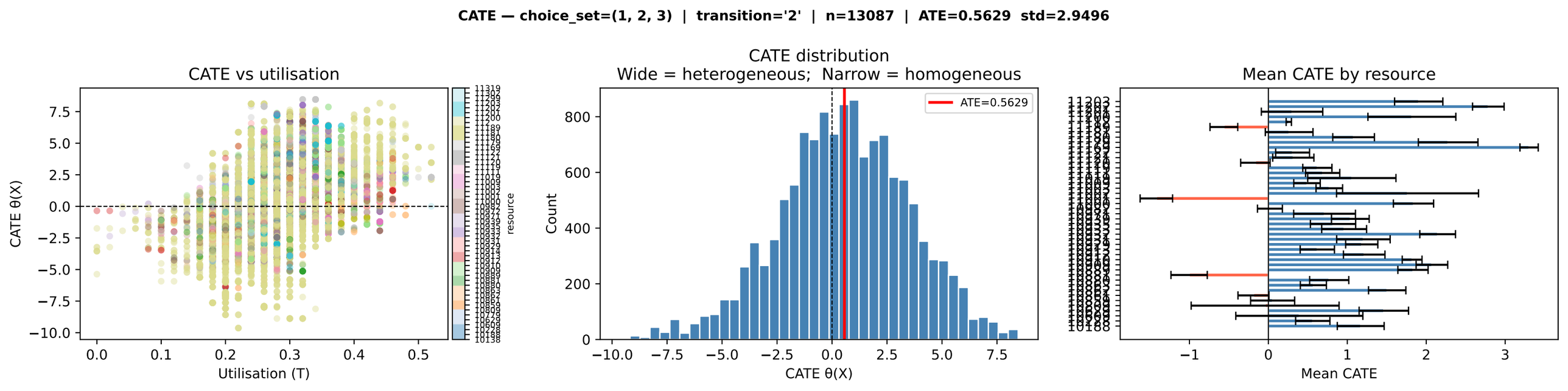
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

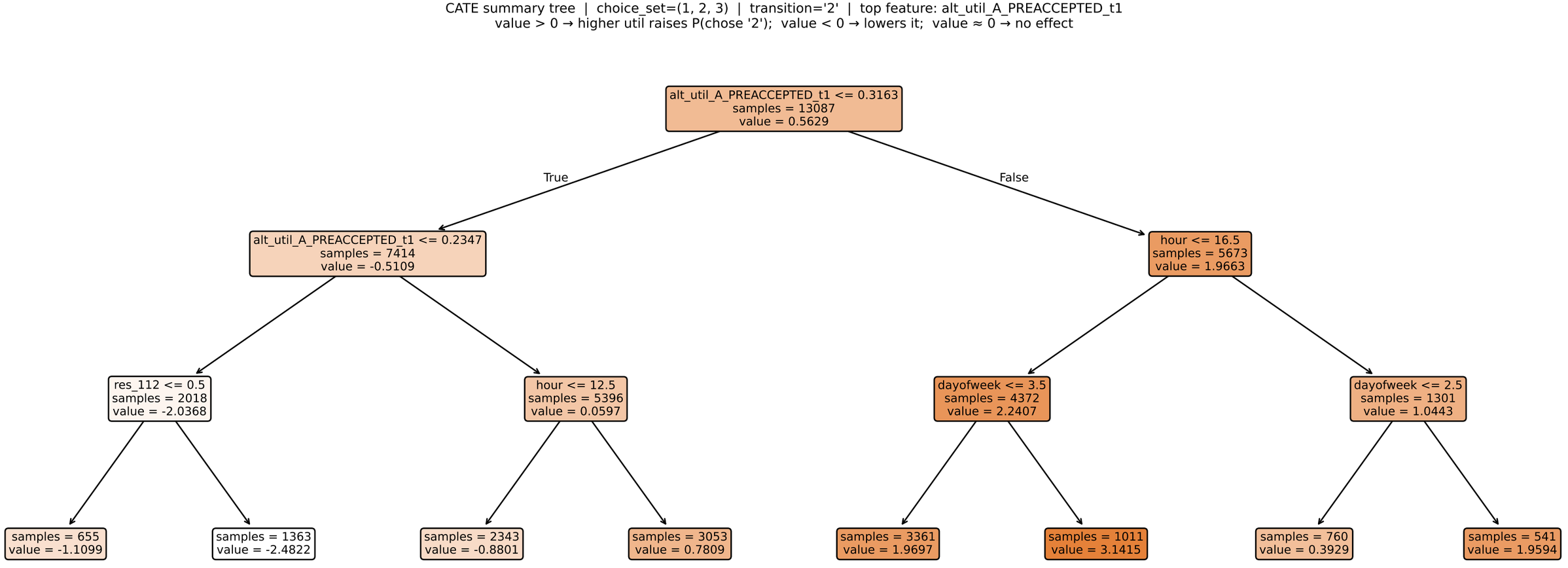
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	1	A_PREACCEPTED_t1	A_CANCELLED_t3, A_DECLINED_t2	0.300921	[-4.7584, 5.3602]	0.9072	False	ATE=+0.3009: 0→1 util ↑ P(t1/A_PREACCEPT
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	2	A_DECLINED_t2	A_CANCELLED_t3, A_PREACCEPTED_t1	0.562914	[-4.7468, 5.8727]	0.8354	False	ATE=+0.5629: 0→1 util ↑ P(t2/A_DECLINED
(1, 2, 3)	('A_PREACCEPTED_t1', 'A_DECLINED_t2', 'A_CANCELLED_t3')	13087	1, 2, 3	3	A_CANCELLED_t3	A_DECLINED_t2, A_PREACCEPTED_t1	-0.000141	[-0.0001, -0.0001]	1.0	False	ATE=-0.0001: 0→1 util ↓ P(t3/A_CANCELLE
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	4	A_DECLINED_t4	A_ACCEPTED_t6, A_CANCELLED_t5	-1.609976	[-6.8660, 3.6460]	0.5483	False	ATE=-1.6100: 0→1 util ↓ P(t4/A_DECLINED_
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	5	A_CANCELLED_t5	A_ACCEPTED_t6, A_DECLINED_t4	0.496191	[-1.4165, 2.4089]	0.6111	False	ATE=+0.4962: 0→1 util ↑ P(t5/A_CANCELLE
(4, 5, 6)	('A_DECLINED_t4', 'A_CANCELLED_t5', 'A_ACCEPTED_t6')	7298	4, 5, 6	6	A_ACCEPTED_t6	A_CANCELLED_t5, A_DECLINED_t4	0.161766	[-5.0686, 5.3921]	0.9517	False	ATE=+0.1618: 0→1 util ↑ P(t6/A_ACCEPTE
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	7	A_DECLINED_t7	A_CANCELLED_t8, A_FINALIZED_t9	-0.003578	[-0.1502, 0.1431]	0.9619	False	ATE=-0.0036: 0→1 util ↓ P(t7/A_DECLINE
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	8	A_CANCELLED_t8	A_DECLINED_t7, A_FINALIZED_t9	-0.020466	[-0.3176, 0.2767]	0.8926	False	ATE=-0.0205: 0→1 util ↓ P(t8/A_CANCELLE
(7, 8, 9)	('A_DECLINED_t7', 'A_CANCELLED_t8', 'A_FINALIZED_t9')	5110	7, 8, 9	9	A_FINALIZED_t9	A_CANCELLED_t8, A_DECLINED_t7	-0.041301	[-0.5116, 0.4290]	0.8633	False	ATE=-0.0413: 0→1 util ↓ P(t9/A_FINALIZE
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	10	A_DECLINED_t10	A_APPROVED_t12, A_CANCELLED_t11	-0.324801	[-2.3833, 1.7337]	0.7571	False	ATE=-0.3248: 0→1 util ↓ P(t10/A_DECLINED
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	11	A_CANCELLED_t11	A_APPROVED_t12, A_DECLINED_t10	-0.218153	[-1.5720, 1.1357]	0.7521	False	ATE=-0.2182: 0→1 util ↓ P(t11/A_CANCELLE
(10, 11, 12)	('A_DECLINED_t10', 'A_CANCELLED_t11', 'A_APPROVED_t12')	3819	10, 11, 12	12	A_APPROVED_t12	A_CANCELLED_t11, A_DECLINED_t10	-0.094982	[-0.7202, 0.5303]	0.7659	False	ATE=-0.0950: 0→1 util ↓ P(t12/A_APPROVE

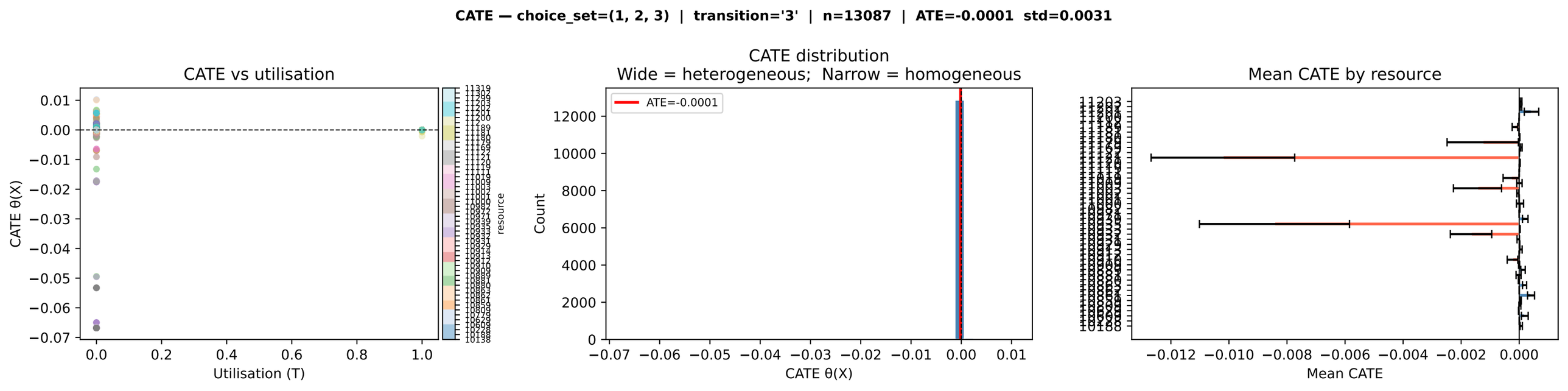


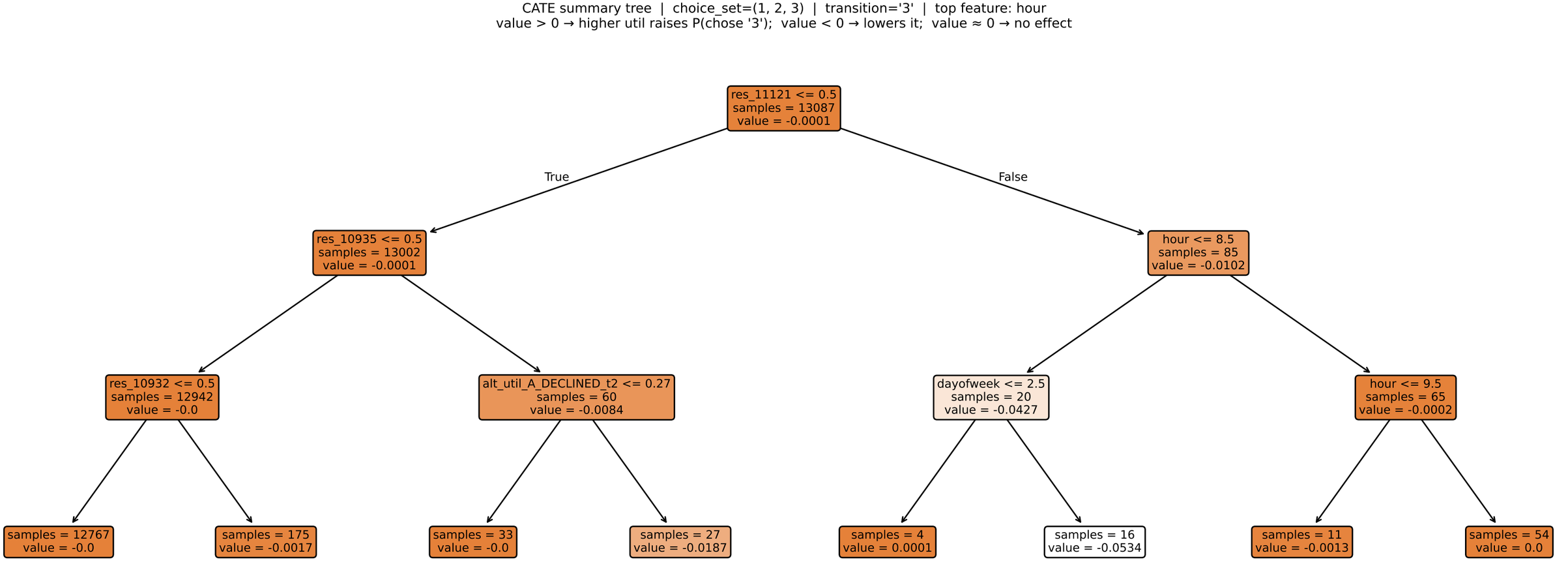

CATE — choice_set=(1, 2, 3) | transition='1' | n=13087 | ATE=0.3009 std=2.7464
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.3009
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

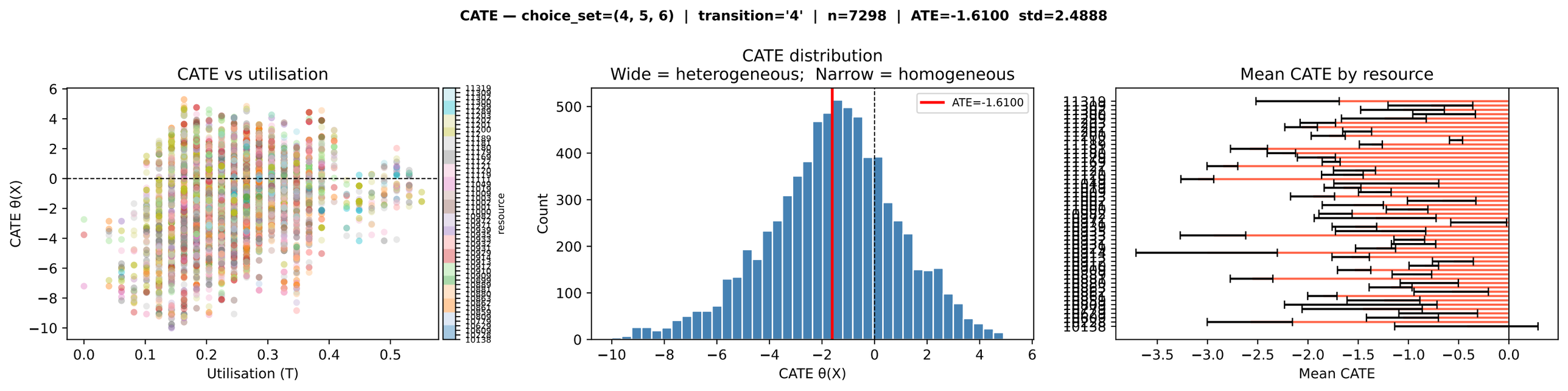

CATE summary tree | choice_set=(1, 2, 3) | transition='1' | top feature: alt_util_A_DECLINED_t2
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_DECLINED_t2 <= 0.23
samples = 13087
value = 0.3009
True
False
res_112 <= 0.5
samples = 2630
value = -1.8949
alt_util_A_DECLINED_t2 <= 0.31
samples = 10457
value = 0.8532
res_11169 <= 0.5
samples = 860
value = -0.9042
alt_util_A_DECLINED_t2 <= 0.21
samples = 1770
value = -2.3762
hour <= 7.5
samples = 5794
value = 0.2327
hour <= 18.5
samples = 4663
value = 1.6241
samples = 814
value = -1.0729
samples = 46
value = 2.0802
samples = 1169
value = -2.7598
samples = 601
value = -1.63
samples = 295
value = -2.3087
samples = 5499
value = 0.3691
samples = 4104
value = 1.8284
samples = 559
value = 0.1241

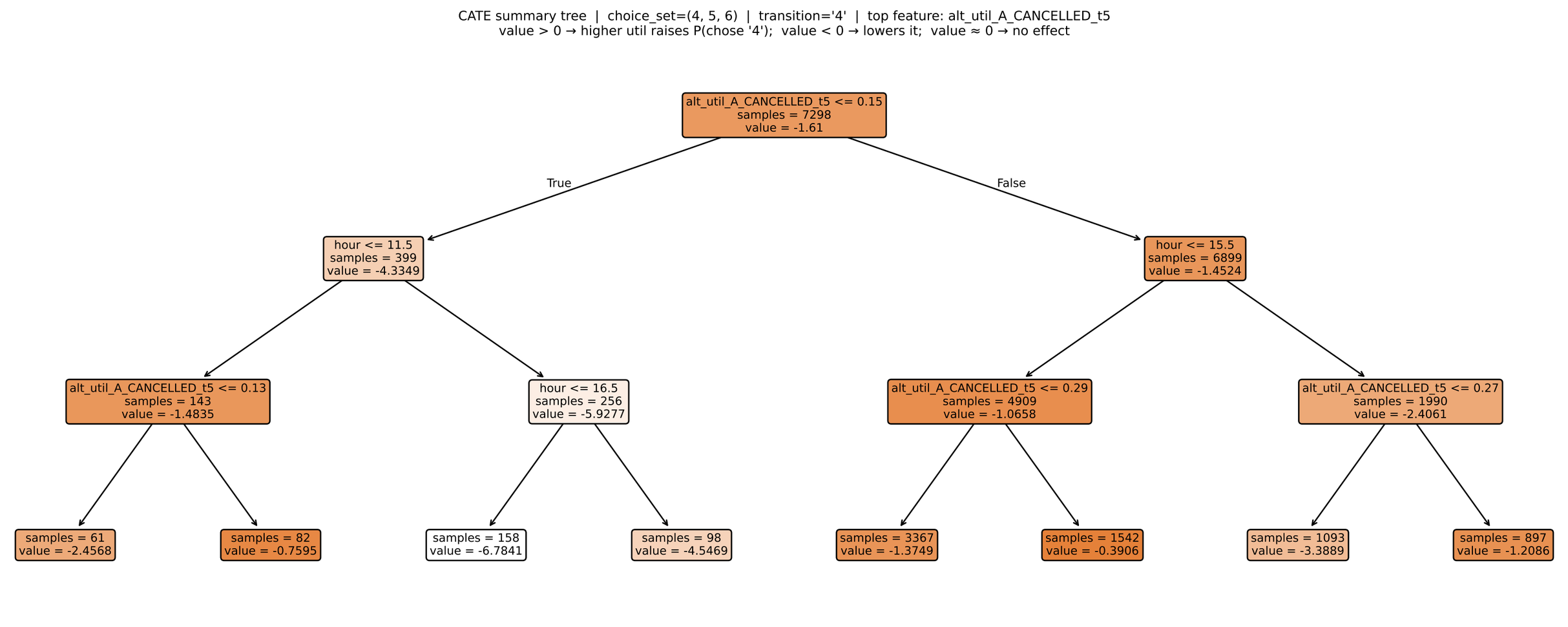

CATE — choice_set=(1, 2, 3) | transition='2' | n=13087 | ATE=0.5629 std=2.9496
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.5629
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

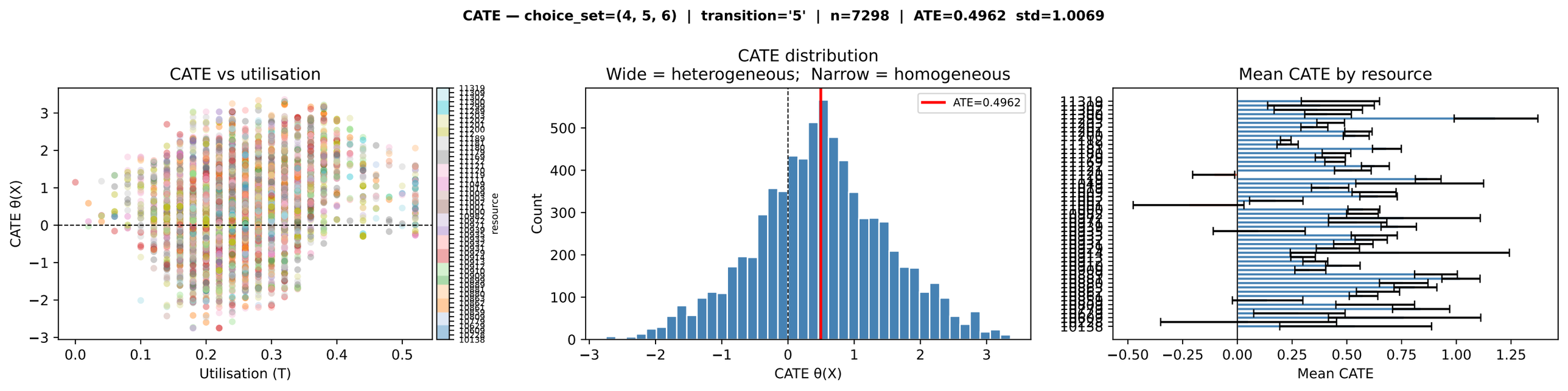

CATE summary tree | choice_set=(1, 2, 3) | transition='2' | top feature: alt_util_A_PREACCEPTED_t1
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_PREACCEPTED_t1 <= 0.3163
samples = 13087
value = 0.5629
True
False
alt_util_A_PREACCEPTED_t1 <= 0.2347
samples = 7414
value = -0.5109
hour <= 16.5
samples = 5673
value = 1.9663
res_112 <= 0.5
samples = 2018
value = -2.0368
hour <= 12.5
samples = 5396
value = 0.0597
dayofweek <= 3.5
samples = 4372
value = 2.2407
dayofweek <= 2.5
samples = 1301
value = 1.0443
samples = 655
value = -1.1099
samples = 1363
value = -2.4822
samples = 2343
value = -0.8801
samples = 3053
value = 0.7809
samples = 3361
value = 1.9697
samples = 1011
value = 3.1415
samples = 760
value = 0.3929
samples = 541
value = 1.9594

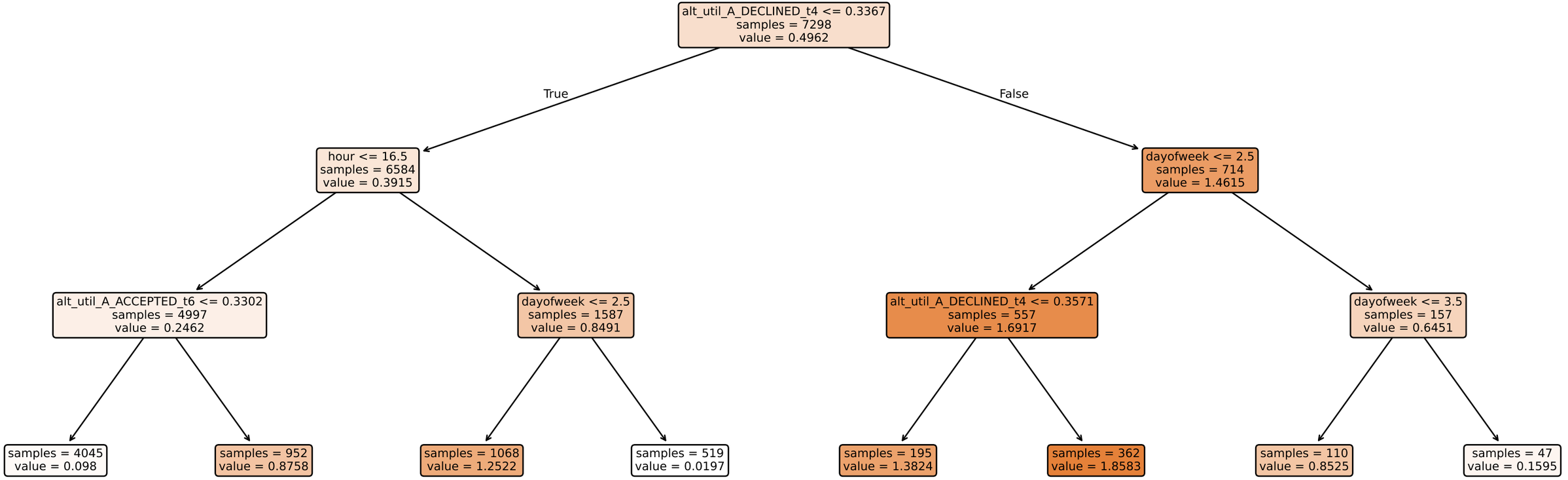

CATE — choice_set=(1, 2, 3) | transition='3' | n=13087 | ATE=-0.0001 std=0.0031
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0001
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

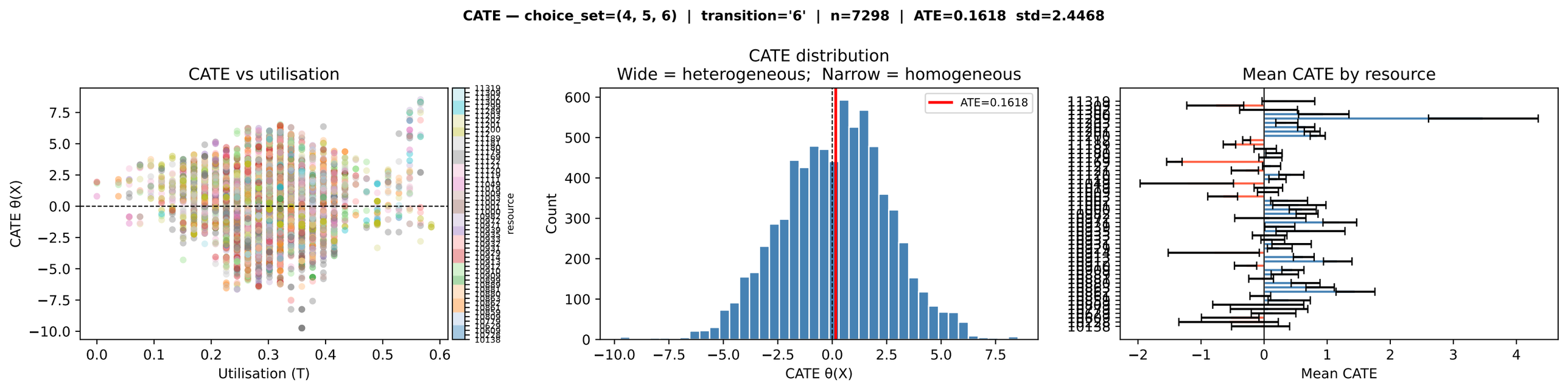

CATE summary tree | choice_set=(1, 2, 3) | transition='3' | top feature: hour
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
res_11121 <= 0.5
samples = 13087
value = -0.0001
True
False
res_10935 <= 0.5
samples = 13002
value = -0.0001
hour <= 8.5
samples = 85
value = -0.0102
res_10932 <= 0.5
samples = 12942
value = -0.0
alt_util_A_DECLINED_t2 <= 0.27
samples = 60
value = -0.0084
dayofweek <= 2.5
samples = 20
value = -0.0427
hour <= 9.5
samples = 65
value = -0.0002
samples = 12767
value = -0.0
samples = 175
value = -0.0017
samples = 33
value = -0.0
samples = 27
value = -0.0187
samples = 4
value = 0.0001
samples = 16
value = -0.0534
samples = 11
value = -0.0013
samples = 54
value = 0.0

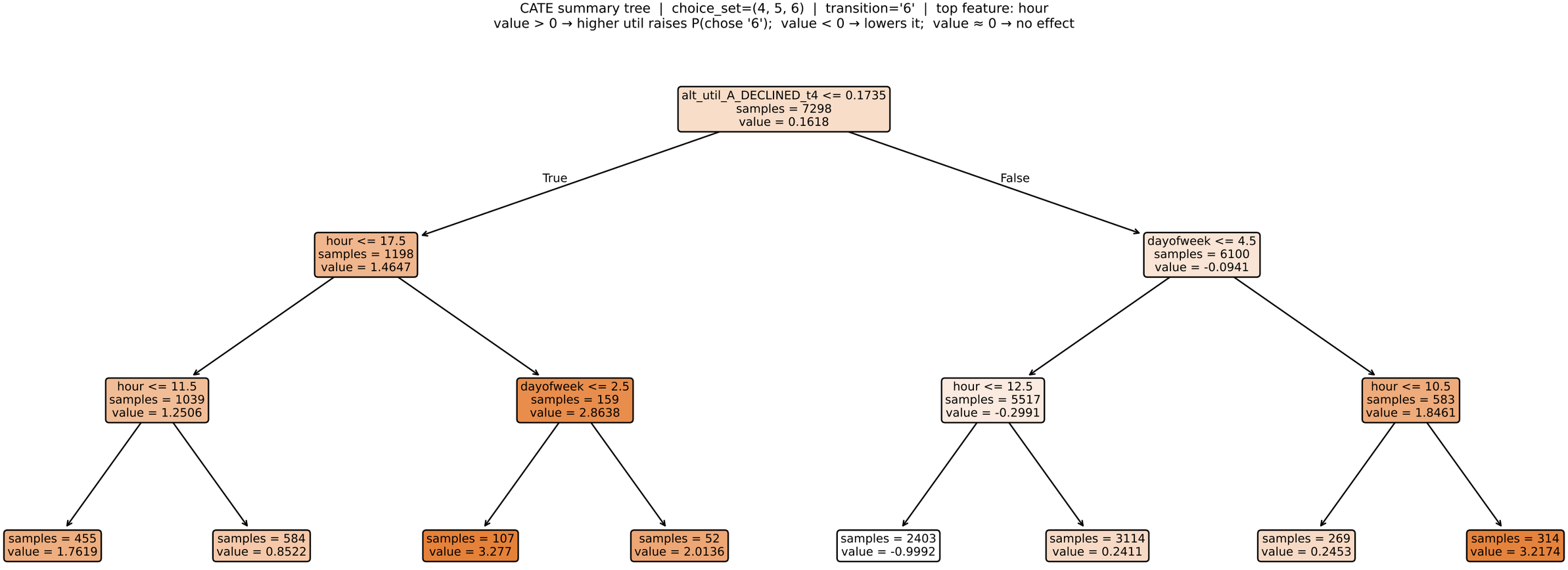

CATE — choice_set=(4, 5, 6) | transition='4' | n=7298 | ATE=-1.6100 std=2.4888
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.6100
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

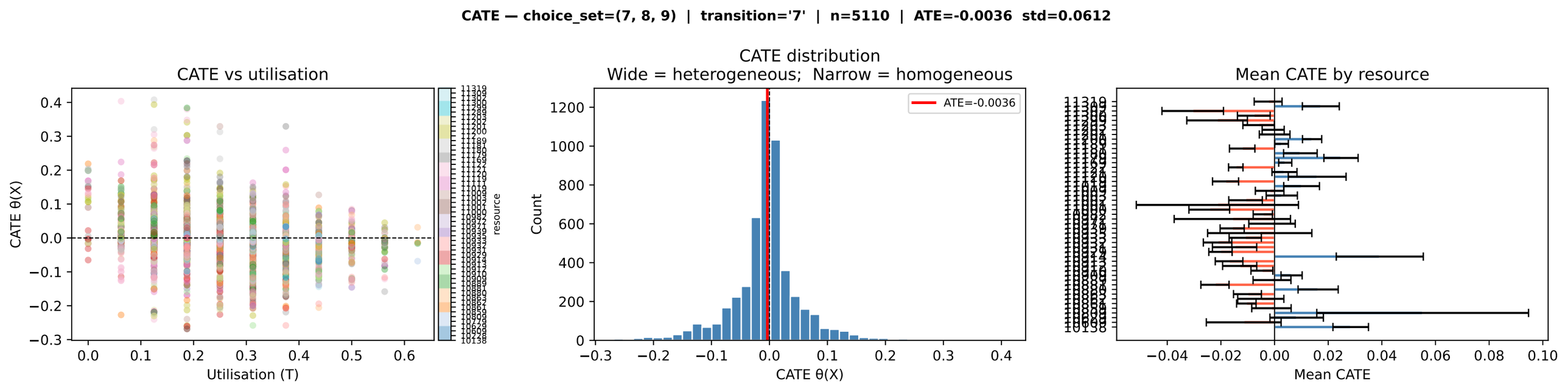

CATE summary tree | choice_set=(4, 5, 6) | transition='4' | top feature: alt_util_A_CANCELLED_t5
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t5 <= 0.15
samples = 7298
value = -1.61
True
False
hour <= 11.5
samples = 399
value = -4.3349
hour <= 15.5
samples = 6899
value = -1.4524
alt_util_A_CANCELLED_t5 <= 0.13
samples = 143
value = -1.4835
hour <= 16.5
samples = 256
value = -5.9277
alt_util_A_CANCELLED_t5 <= 0.29
samples = 4909
value = -1.0658
alt_util_A_CANCELLED_t5 <= 0.27
samples = 1990
value = -2.4061
samples = 61
value = -2.4568
samples = 82
value = -0.7595
samples = 158
value = -6.7841
samples = 98
value = -4.5469
samples = 3367
value = -1.3749
samples = 1542
value = -0.3906
samples = 1093
value = -3.3889
samples = 897
value = -1.2086

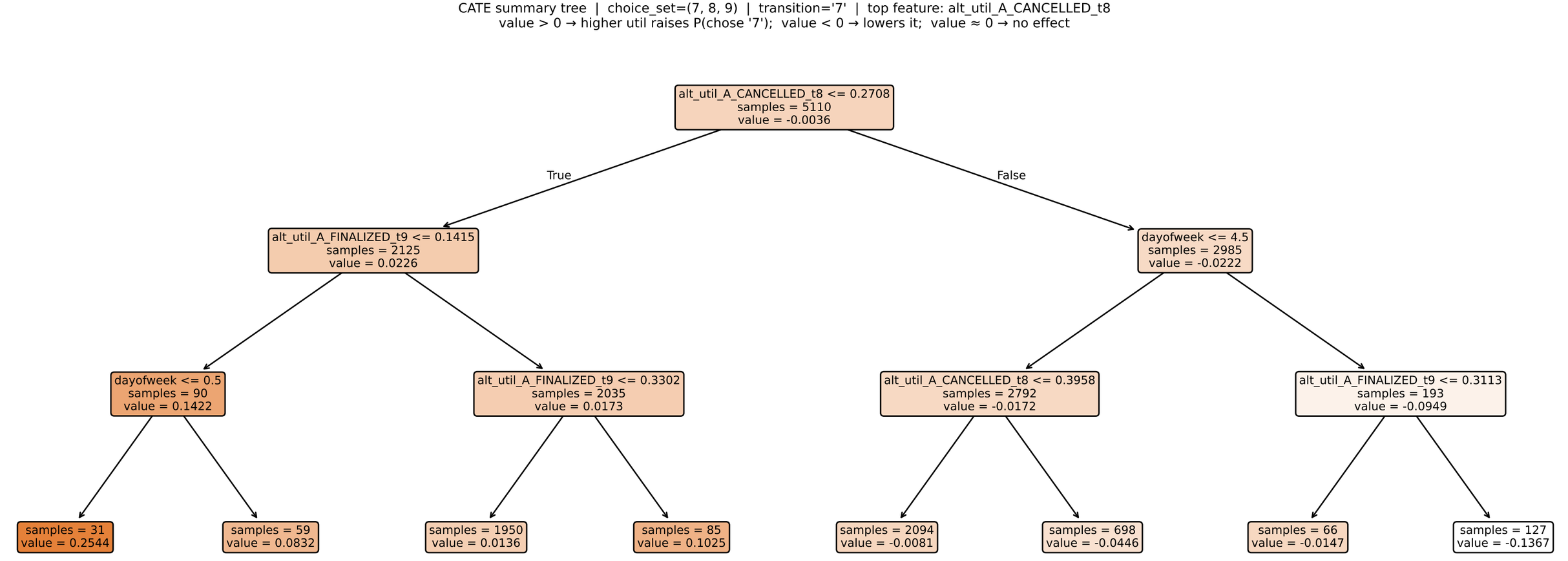

CATE — choice_set=(4, 5, 6) | transition='5' | n=7298 | ATE=0.4962 std=1.0069
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=0.4962
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

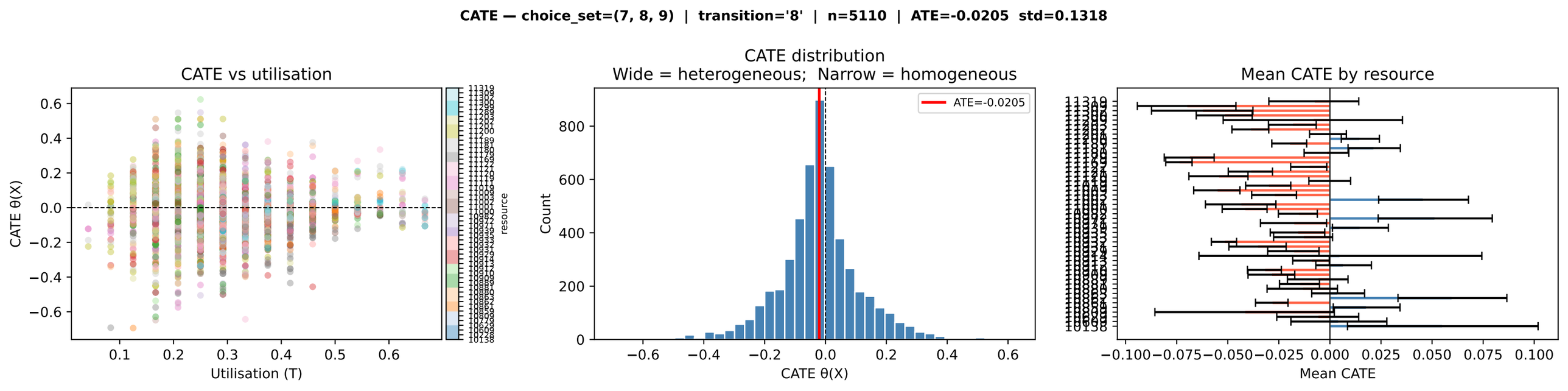

CATE summary tree | choice_set=(4, 5, 6) | transition='5' | top feature: alt_util_A_DECLINED_t4
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_DECLINED_t4 <= 0.3367
samples = 7298
value = 0.4962
True
False
hour <= 16.5
samples = 6584
value = 0.3915
dayofweek <= 2.5
samples = 714
value = 1.4615
alt_util_A_ACCEPTED_t6 <= 0.3302
samples = 4997
value = 0.2462
dayofweek <= 2.5
samples = 1587
value = 0.8491
alt_util_A_DECLINED_t4 <= 0.3571
samples = 557
value = 1.6917
dayofweek <= 3.5
samples = 157
value = 0.6451
samples = 4045
value = 0.098
samples = 952
value = 0.8758
samples = 1068
value = 1.2522
samples = 519
value = 0.0197
samples = 195
value = 1.3824
samples = 362
value = 1.8583
samples = 110
value = 0.8525
samples = 47
value = 0.1595

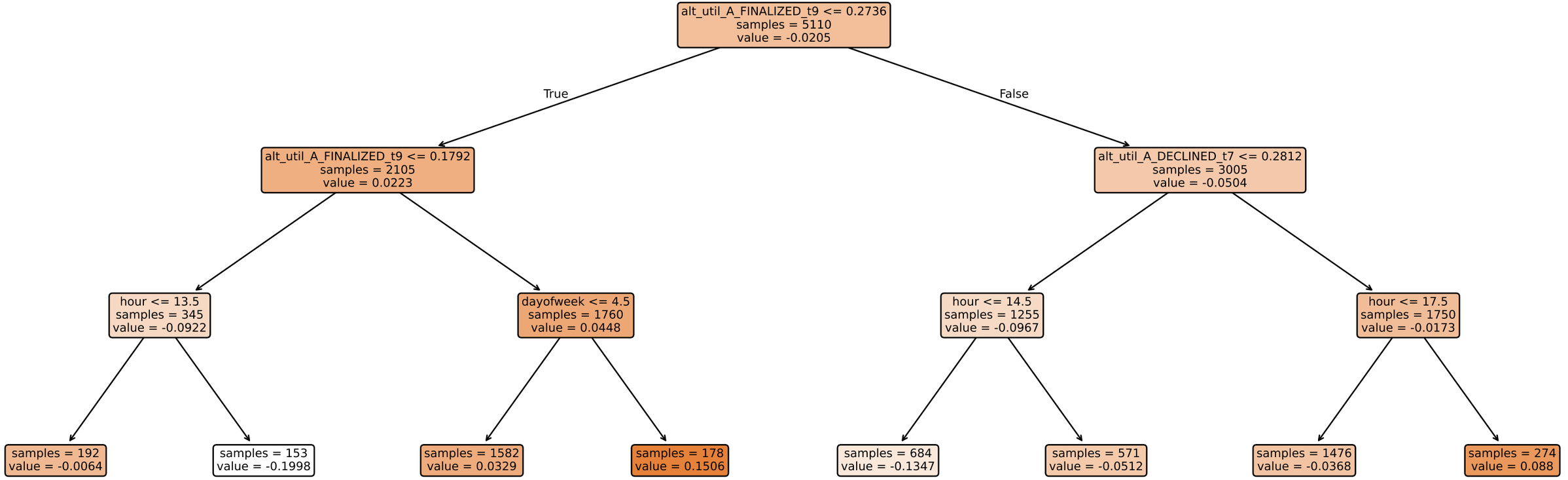

CATE — choice_set=(4, 5, 6) | transition='6' | n=7298 | ATE=0.1618 std=2.4468
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=0.1618
Mean CATE by resource
Mean CATE

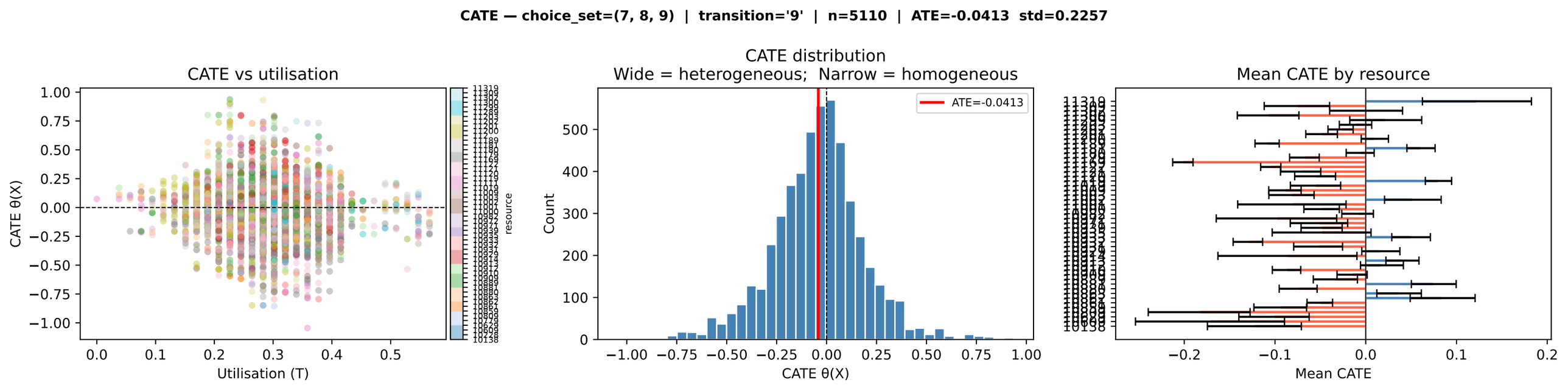

CATE summary tree | choice_set=(4, 5, 6) | transition='6' | top feature: hour
value > 0 → higher util raises P(chose '6'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_DECLINED_t4 <= 0.1735
samples = 7298
value = 0.1618
True
False
hour <= 17.5
samples = 1198
value = 1.4647
dayofweek <= 4.5
samples = 6100
value = -0.0941
hour <= 11.5
samples = 1039
value = 1.2506
dayofweek <= 2.5
samples = 159
value = 2.8638
hour <= 12.5
samples = 5517
value = -0.2991
hour <= 10.5
samples = 583
value = 1.8461
samples = 455
value = 1.7619
samples = 584
value = 0.8522
samples = 107
value = 3.277
samples = 52
value = 2.0136
samples = 2403
value = -0.9992
samples = 3114
value = 0.2411
samples = 269
value = 0.2453
samples = 314
value = 3.2174

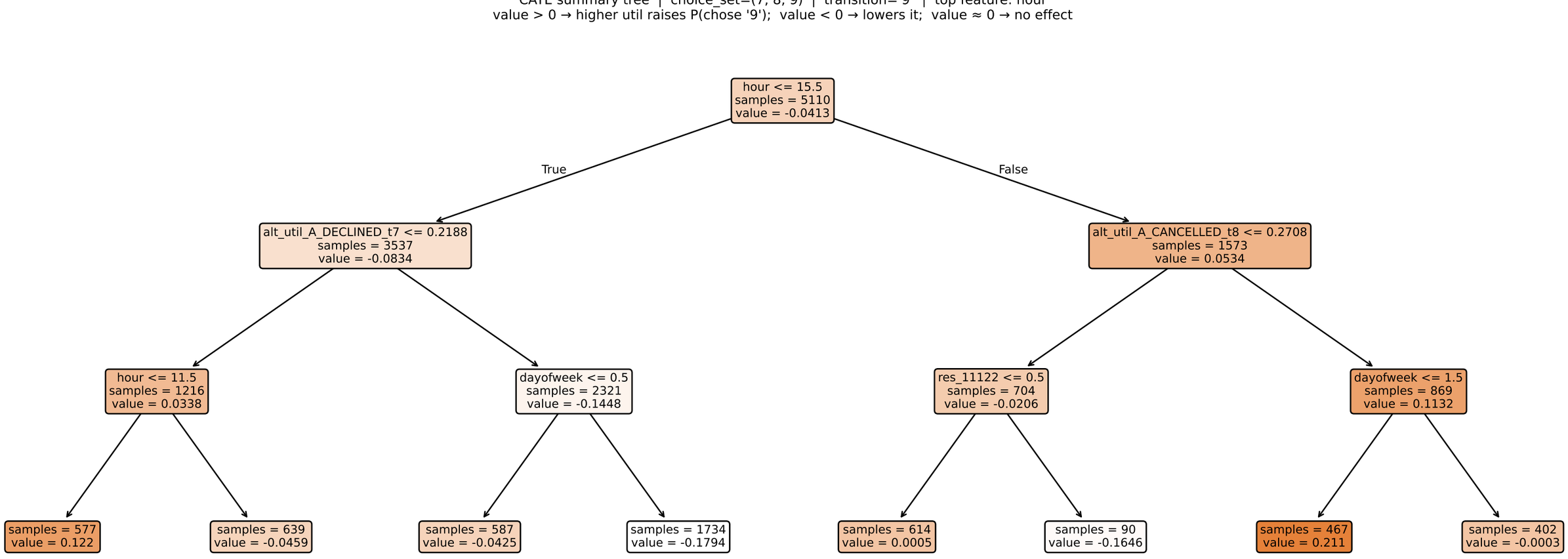

CATE — choice_set=(7, 8, 9) | transition='7' | n=5110 | ATE=-0.0036 std=0.0612
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0036
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

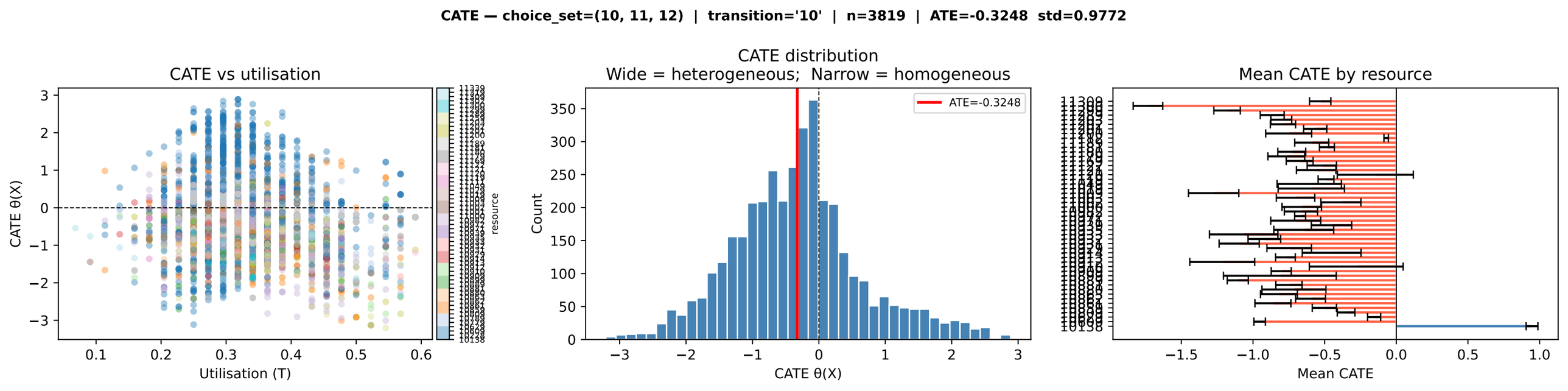

CATE summary tree | choice_set=(7, 8, 9) | transition='7' | top feature: alt_util_A_CANCELLED_t8
value > 0 → higher util raises P(chose '7'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t8 <= 0.2708
samples = 5110
value = -0.0036
True
False
alt_util_A_FINALIZED_t9 <= 0.1415
samples = 2125
value = 0.0226
dayofweek <= 4.5
samples = 2985
value = -0.0222
dayofweek <= 0.5
samples = 90
value = 0.1422
alt_util_A_FINALIZED_t9 <= 0.3302
samples = 2035
value = 0.0173
alt_util_A_CANCELLED_t8 <= 0.3958
samples = 2792
value = -0.0172
alt_util_A_FINALIZED_t9 <= 0.3113
samples = 193
value = -0.0949
samples = 31
value = 0.2544
samples = 59
value = 0.0832
samples = 1950
value = 0.0136
samples = 85
value = 0.1025
samples = 2094
value = -0.0081
samples = 698
value = -0.0446
samples = 66
value = -0.0147
samples = 127
value = -0.1367

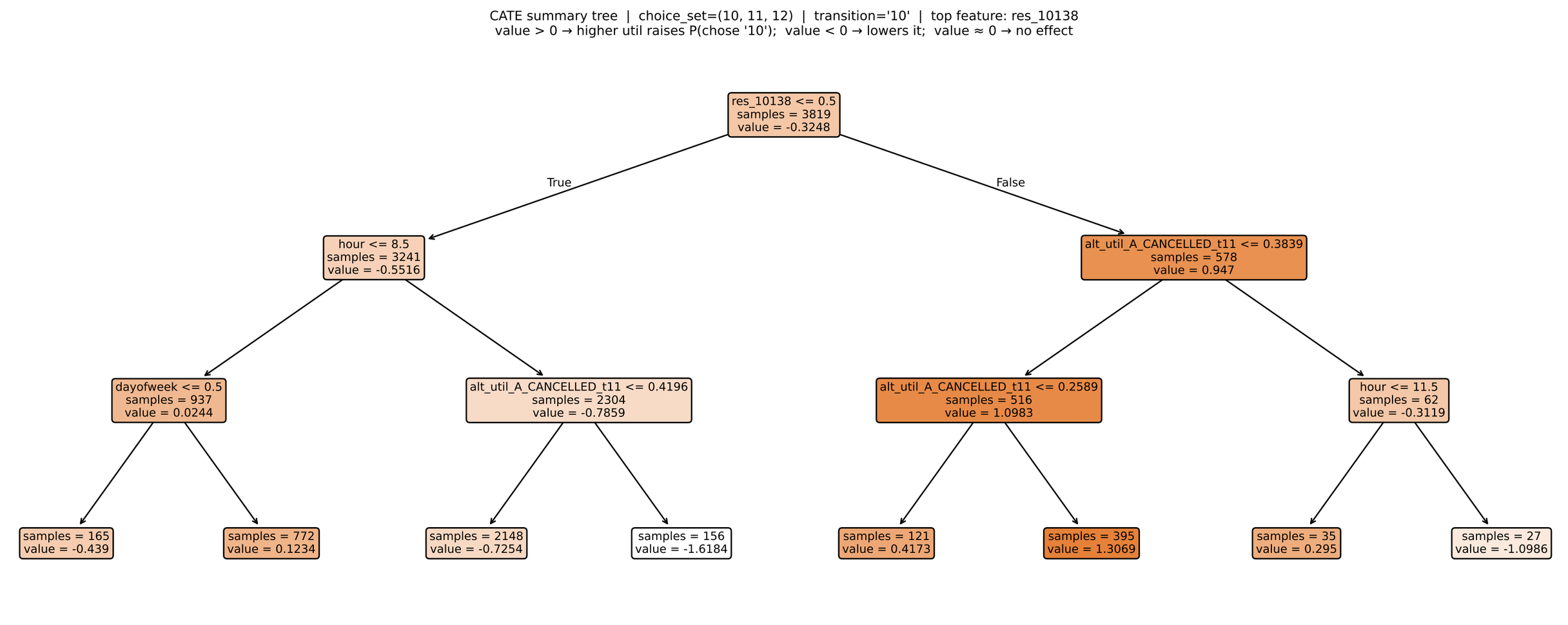

CATE — choice_set=(7, 8, 9) | transition='8' | n=5110 | ATE=-0.0205 std=0.1318
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0205
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

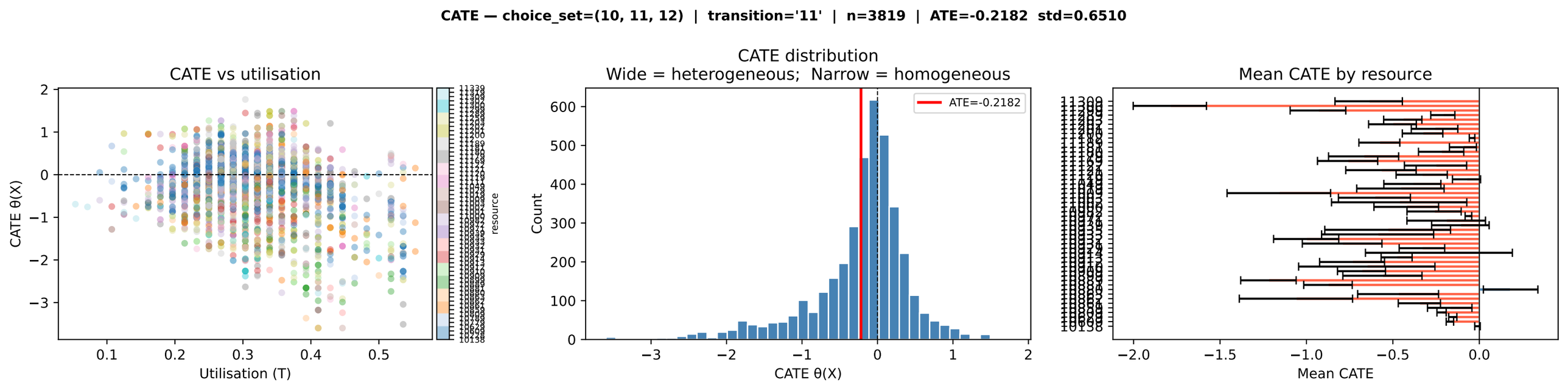

CATE summary tree | choice_set=(7, 8, 9) | transition='8' | top feature: alt_util_A_FINALIZED_t9
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_FINALIZED_t9 <= 0.2736
samples = 5110
value = -0.0205
True
False
alt_util_A_FINALIZED_t9 <= 0.1792
samples = 2105
value = 0.0223
alt_util_A_DECLINED_t7 <= 0.2812
samples = 3005
value = -0.0504
hour <= 13.5
samples = 345
value = -0.0922
dayofweek <= 4.5
samples = 1760
value = 0.0448
hour <= 14.5
samples = 1255
value = -0.0967
hour <= 17.5
samples = 1750
value = -0.0173
samples = 192
value = -0.0064
samples = 153
value = -0.1998
samples = 1582
value = 0.0329
samples = 178
value = 0.1506
samples = 684
value = -0.1347
samples = 571
value = -0.0512
samples = 1476
value = -0.0368
samples = 274
value = 0.088

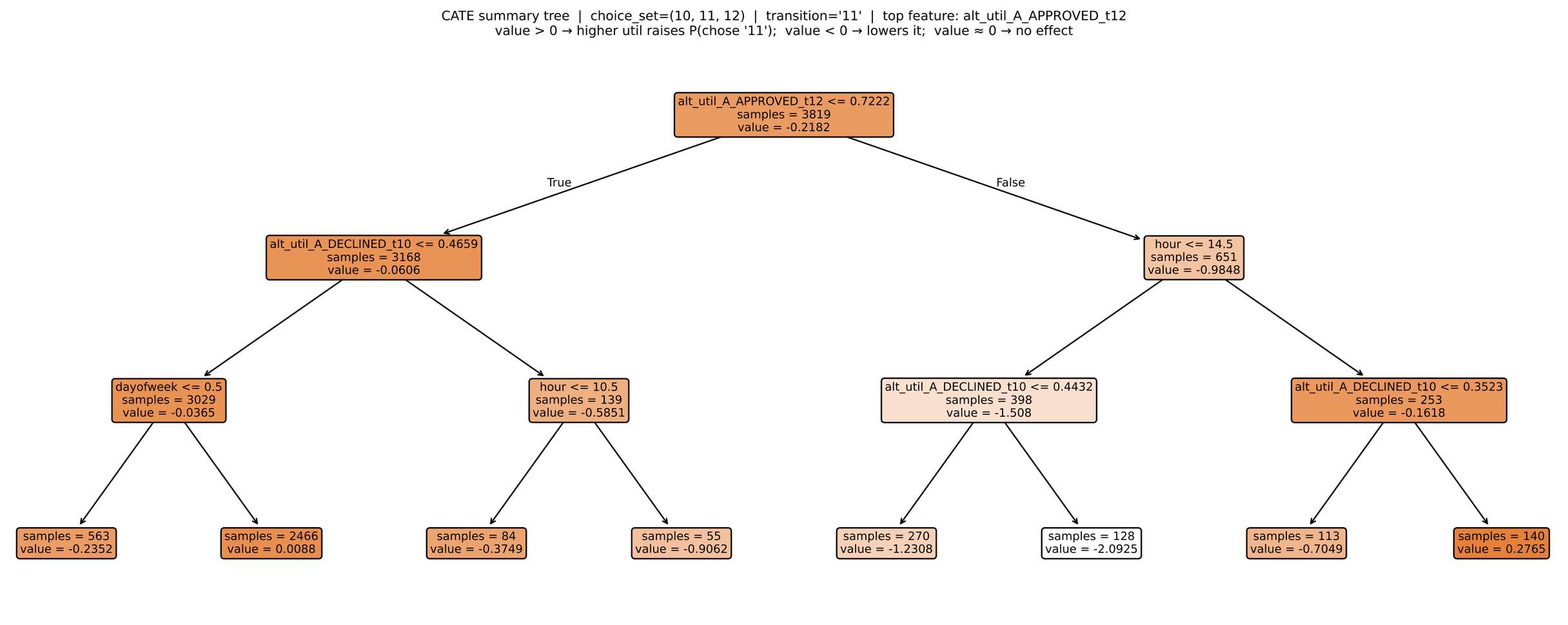

CATE — choice_set=(7, 8, 9) | transition='9' | n=5110 | ATE=-0.0413 std=0.2257
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0413
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

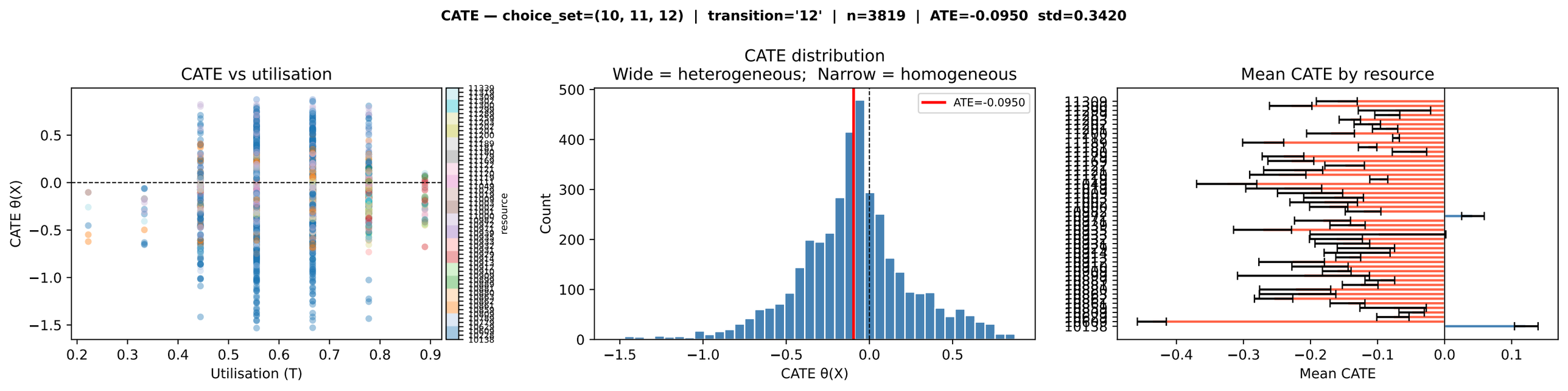

CATE summary tree | choice_set=(7, 8, 9) | transition='9' | top feature: hour
value > 0 → higher util raises P(chose '9'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 15.5
samples = 5110
value = -0.0413
True
False
alt_util_A_DECLINED_t7 <= 0.2188
samples = 3537
value = -0.0834
alt_util_A_CANCELLED_t8 <= 0.2708
samples = 1573
value = 0.0534
hour <= 11.5
samples = 1216
value = 0.0338
dayofweek <= 0.5
samples = 2321
value = -0.1448
res_11122 <= 0.5
samples = 704
value = -0.0206
dayofweek <= 1.5
samples = 869
value = 0.1132
samples = 577
value = 0.122
samples = 639
value = -0.0459
samples = 587
value = -0.0425
samples = 1734
value = -0.1794
samples = 614
value = 0.0005
samples = 90
value = -0.1646
samples = 467
value = 0.211
samples = 402
value = -0.0003


CATE — choice_set=(10, 11, 12) | transition='10' | n=3819 | ATE=-0.3248 std=0.9772
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.3248
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(10, 11, 12) | transition='10' | top feature: res_10138
value > 0 → higher util raises P(chose '10'); value < 0 → lowers it; value ≈ 0 → no effect
res_10138 <= 0.5
samples = 3819
value = -0.3248
True
False
hour <= 8.5
samples = 3241
value = -0.5516
alt_util_A_CANCELLED_t11 <= 0.3839
samples = 578
value = 0.947
dayofweek <= 0.5
samples = 937
value = 0.0244
alt_util_A_CANCELLED_t11 <= 0.4196
samples = 2304
value = -0.7859
alt_util_A_CANCELLED_t11 <= 0.2589
samples = 516
value = 1.0983
hour <= 11.5
samples = 62
value = -0.3119
samples = 165
value = -0.439
samples = 772
value = 0.1234
samples = 2148
value = -0.7254
samples = 156
value = -1.6184
samples = 121
value = 0.4173
samples = 395
value = 1.3069
samples = 35
value = 0.295
samples = 27
value = -1.0986


CATE — choice_set=(10, 11, 12) | transition='11' | n=3819 | ATE=-0.2182 std=0.6510
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
Count
CATE θ(X)
ATE=-0.2182
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(10, 11, 12) | transition='11' | top feature: alt_util_A_APPROVED_t12
value > 0 → higher util raises P(chose '11'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_APPROVED_t12 <= 0.7222
samples = 3819
value = -0.2182
True
False
alt_util_A_DECLINED_t10 <= 0.4659
samples = 3168
value = -0.0606
hour <= 14.5
samples = 651
value = -0.9848
dayofweek <= 0.5
samples = 3029
value = -0.0365
hour <= 10.5
samples = 139
value = -0.5851
alt_util_A_DECLINED_t10 <= 0.4432
samples = 398
value = -1.508
alt_util_A_DECLINED_t10 <= 0.3523
samples = 253
value = -0.1618
samples = 563
value = -0.2352
samples = 2466
value = 0.0088
samples = 84
value = -0.3749
samples = 55
value = -0.9062
samples = 270
value = -1.2308
samples = 128
value = -2.0925
samples = 113
value = -0.7049
samples = 140
value = 0.2765


CATE — choice_set=(10, 11, 12) | transition='12' | n=3819 | ATE=-0.0950 std=0.3420
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0950
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

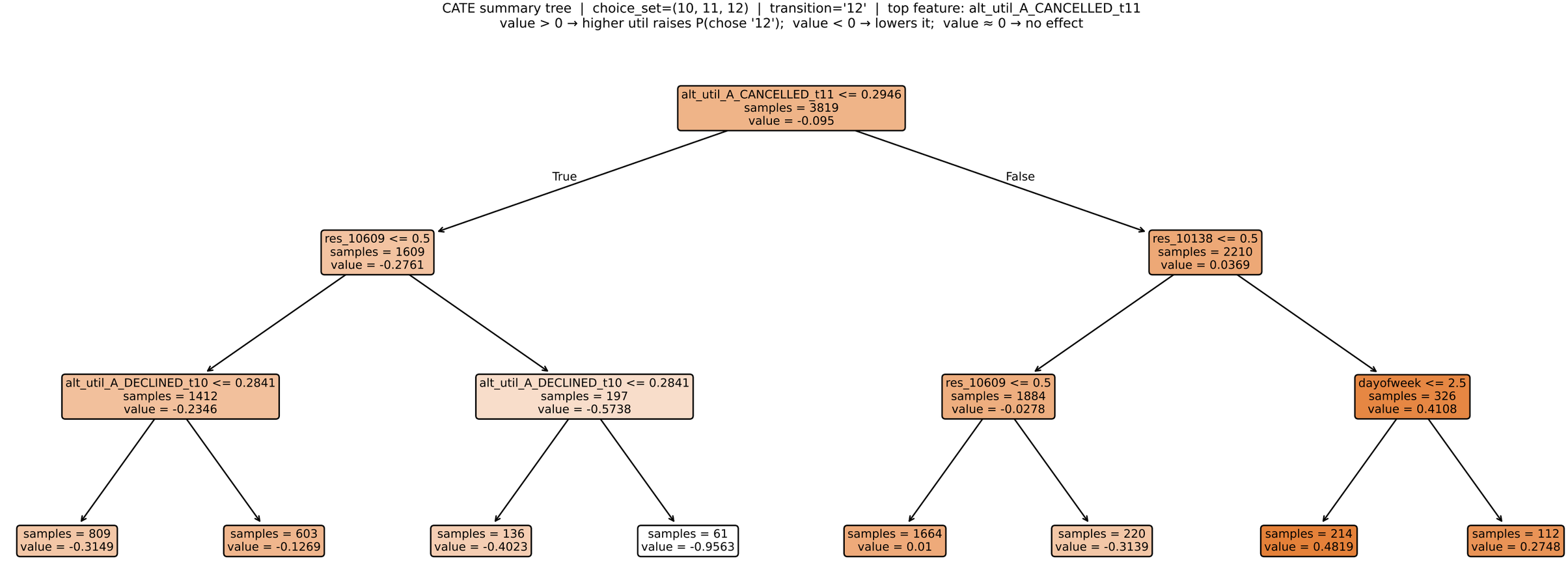

CATE summary tree | choice_set=(10, 11, 12) | transition='12' | top feature: alt_util_A_CANCELLED_t11
value > 0 → higher util raises P(chose '12'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_A_CANCELLED_t11 <= 0.2946
samples = 3819
value = -0.095
True
False
res_10609 <= 0.5
samples = 1609
value = -0.2761
res_10138 <= 0.5
samples = 2210
value = 0.0369
alt_util_A_DECLINED_t10 <= 0.2841
samples = 1412
value = -0.2346
alt_util_A_DECLINED_t10 <= 0.2841
samples = 197
value = -0.5738
res_10609 <= 0.5
samples = 1884
value = -0.0278
dayofweek <= 2.5
samples = 326
value = 0.4108
samples = 809
value = -0.3149
samples = 603
value = -0.1269
samples = 136
value = -0.4023
samples = 61
value = -0.9563
samples = 1664
value = 0.01
samples = 220
value = -0.3139
samples = 214
value = 0.4819
samples = 112
value = 0.2748

Table 1. Participant demographics.

Characteristic	Overall	Control	Intervention	p-value
Age, mean (SD)	45.2 (12.8)	44.9 (12.6)	45.5 (13.0)	0.412
Female, n (%)	1,284 (52.1)	641 (51.8)	643 (52.4)	0.781
Body mass index, mean (SD)	27.4 (4.6)	27.3 (4.5)	27.5 (4.7)	0.233
Smoker, n (%)	312 (12.7)	158 (12.8)	154 (12.5)	0.842
Hypertension, n (%)	498 (20.2)	251 (20.3)	247 (20.1)	0.901
Diabetes, n (%)	187 (7.6)	92 (7.4)	95 (7.7)	0.756
Baseline systolic BP, mean (SD)	128.6 (15.2)	128.9 (15.4)	128.3 (15.0)	0.318
Baseline diastolic BP, mean (SD)	82.1 (9.8)	82.3 (9.9)	81.9 (9.7)	0.214
Completed follow-up, n (%)	2,104 (85.3)	1,041 (84.1)	1,063 (86.4)	0.118

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(1, 2, 3)	1	13087	0.300921	2.746353	-8.547345	-1.512017	0.344175	2.124421	7.96925	55.4	44.6	alt_util_A_DECLINED_t2
(1, 2, 3)	2	13087	0.562914	2.949626	-9.634255	-1.342523	0.600182	2.573298	8.462791	57.6	42.4	alt_util_A_PREACCEPTED_t1
(1, 2, 3)	3	13087	-0.000141	0.003064	-0.066833	-0.0	0.0	0.0	0.01028	52.3	47.7	hour
(4, 5, 6)	4	7298	-1.609976	2.488812	-10.00815	-3.114257	-1.475619	0.004683	5.293212	25.1	74.9	alt_util_A_CANCELLED_t5
(4, 5, 6)	5	7298	0.496191	1.006948	-2.749754	-0.138677	0.49359	1.139524	3.362331	70.4	29.6	alt_util_A_DECLINED_t4
(4, 5, 6)	6	7298	0.161766	2.44682	-9.741597	-1.523507	0.311212	1.809997	8.538737	54.0	46.0	hour
(7, 8, 9)	7	5110	-0.003578	0.061172	-0.268412	-0.024225	-0.001878	0.016376	0.407951	46.4	53.6	alt_util_A_CANCELLED_t8
(7, 8, 9)	8	5110	-0.020466	0.1318	-0.693809	-0.083197	-0.020996	0.03879	0.623405	38.9	61.1	alt_util_A_FINALIZED_t9
(7, 8, 9)	9	5110	-0.041301	0.225684	-1.045329	-0.174266	-0.03307	0.083996	0.936716	42.6	57.4	hour
(10, 11, 12)	10	3819	-0.324801	0.977183	-3.206836	-0.956931	-0.33267	0.160087	2.886905	31.2	68.8	res_10138
(10, 11, 12)	11	3819	-0.218153	0.651031	-3.598571	-0.419562	-0.089126	0.139598	1.764097	39.4	60.6	alt_util_A_APPROVED_t12
(10, 11, 12)	12	3819	-0.094982	0.341982	-1.532574	-0.284881	-0.087512	0.072498	0.875907	33.4	66.6	alt_util_A_CANCELLED_t11

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md